INCH-POUND

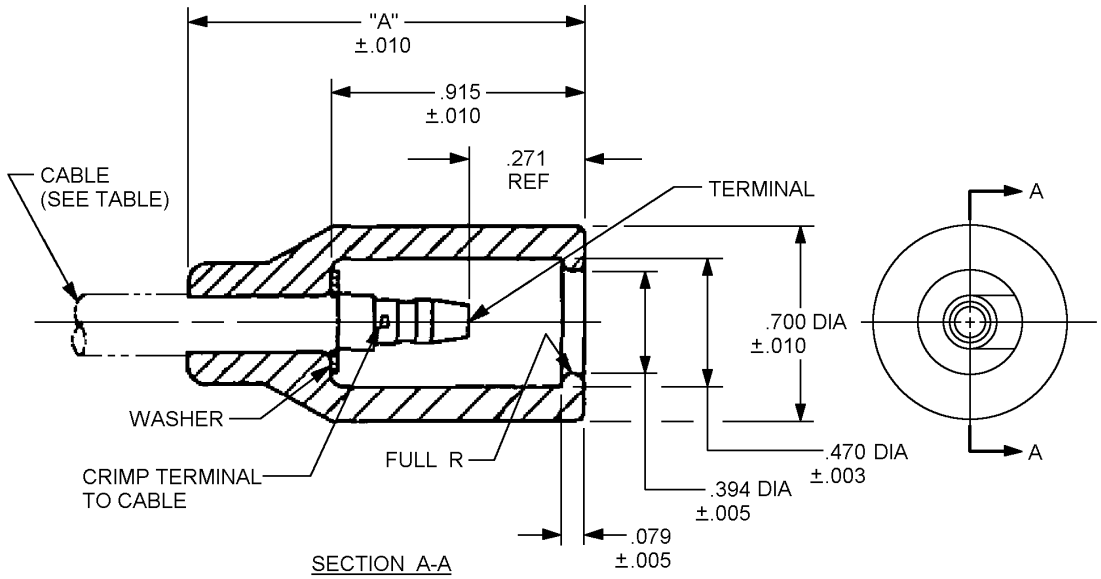
MS27142E
1 March 2018
SUPERSEDING
MS27142D
17 June 2003

DETAIL SPECIFICATION SHEET

CONNECTOR, PLUG, ELECTRICAL, PIN CONTACT, NO. 12, 14 AND 16 AWG, WATERPROOF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet.



Inches	mm	Inches	mm	Inches	mm	Inches	mm
.003	0.076	.079	2.00	.271	6.88	.915	23.24
.005	0.127	.135	3.43	.394	10.01	1.328	33.73
.010	0.254	.160	4.06	.470	11.94	1.421	36.09
.015	0.381	.235	5.96	.700	17.78		

FIGURE 1. Configuration and dimensions.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime - VAI, P.O. Box 3990, Columbus OH 43218-3990 or emailed to CircularConnector@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A

FSC 5935

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:16Z
Check the source to verify that this is the current version before use.

MS dash no.	Reference drawing no. (assembly)	Size	Shell number	Washer number	Terminal number PIN per A-A-52536	A	Cable diameter	AWG size	Cable MIL-DTL-13486 PIN (see application note)
-1	8338571	1	8338572	8338573	AA52536-1	1.421	.235 ± .010	12	M13486/1-7
-2	8338565	2	8338566	8338567	AA52536-2	1.328	.160 ± .010	14	M13486/1-5
-3	None	3	8724495	8724497	AA52536-3	1.328	.135 ± .015	16	M13486/1-3

FIGURE 1. Configuration and dimensions - Continued.

REQUIREMENTS:

Material:

Washer: Brass alloy C24000, C26000 or C26800, temper H02, in accordance with ASTM-B36/B36M.

Shell: Rubber in accordance with ASTM-D2000, M3BC610A14C12EO34F19.

Mating connectors:

MS27142-1 (size 1) mates with connector PIN MS27143-1 in accordance with MS27143.
MS27142-2 (size 2) mates with connector PIN MS27144-1 in accordance with MS27144, style 1, size 1.
MS27142-3 (size 3) mates with connector PIN MS27144-2 in accordance with MS27144, style 1, size 2.

Electrical requirements:

Voltage rating: 30 volts DC.

Current rating:

AWG size	Current (AMPS)
16	13
14	17
12	23

Voltage drop: When connections are coupled, the voltage drop shall not exceed 20 millivolts and shall meet the requirements outlined in MIL-STD-202-307.

Dielectric strength: When connectors are coupled, the assembly shall meet the requirements of the dielectric strength test (1000 volts) outlined in MIL-STD-202-302.

Salt spray **c**orrosion test: When connectors are coupled, the assembly shall meet the salt spray corrosion test outlined in MIL-STD-202-101 (5% saline solution, 200 hours duration).

Waterproof requirements: When connectors are coupled, the assembly shall be waterproof. Refer to MIL-HDBK-1184, or equivalent, for guidance on waterproof testing.

Pull test: The terminal crimped to the cable shall withstand a pull test of 90 pounds (40 kilograms) on size 1, 60 pounds (27 kilograms) on size 2, and 40 pounds (18 kilograms) on size 3.

Separation force: The force required to uncouple the connectors shall not exceed 50 percent of the values expressed in the pull test above at -55°C and at 85°C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:16Z
Check the source to verify that this is the current version before use.

Part or Identifying Number (PIN): MS27142 - (dash number as shown on figure 1).

Referenced drawings may be obtained from the U.S. Army Tank Automotive Command, Warren, MI., 48090.

APPLICATION NOTES:

(These application notes contain information of a general or explanatory nature and are provided for information only.)

Waterproof seal, cable selection and assembly: Recommend that only MIL-DTL-13486/1 cable be used for military applications in order to obtain a waterproof assembly. Other types of cable with smaller outer diameters have been found to be too small to provide a waterproof seal. In addition, to aid in assembling the cable to the connector, the user may apply isopropyl alcohol as an assembly lubricant. Per Army pamphlet AMCP 706-360, a silicone compound in accordance with SAE-AS8660 may be applied as an alternative assembly lubricant.

Nomenclature: For this specification, connector nomenclature is based on the gender of the connector shell. Single wire connectors with pin contacts specified by this document are commonly referred to as “female connectors”. The associated MS27143 and MS27144 mating connectors with socket contacts are commonly referred to as “male connectors”.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. This document references the following:

MIL-DTL-13486
MIL-DTL-13486/1
MS27143
MS27144
MIL-STD-202-101
MIL-STD-202-302
MIL-STD-202-307
MIL-HDBK-1184
ASTM-B36/B36M
ASTM-D2000
A-A-52536
SAE-AS8660
AMCP 706-360
TACOM DWG 8338565
TACOM DWG 8338571

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:16Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - AT
Air Force – 85
Navy - MC
DLA – CC

Preparing activity:
DLA - CC

(Project 5935-2018-042)

Review activities:
Army - CR4
Air Force –99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T15:16Z
Check the source to verify that this is the current version before use.